

Bluestock Fintech

Capstone Project

Mutual Fund Analytics Platform

Final Report

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1. Executive Summary

This project delivers an end-to-end data analytics platform for the Indian mutual fund industry, built for Bluestock Fintech as a 5-day individual capstone. The platform covers data ingestion from real AMFI and mfapi.in sources, data cleaning, SQL database design, exploratory data analysis, fund performance analytics, and an interactive Power BI dashboard.

The analysis spans 40 real mutual fund schemes across 10 major Asset Management Companies (AMCs), covering 4.5 years of NAV history (Jan 2022 to May 2026), over 32,000 investor transactions, and quarterly AUM data for the top 10 fund houses in India.

Key outcomes include a cleaned and validated SQLite database with a 6-table star schema, 13+ exploratory charts, a composite fund scorecard ranking all 40 schemes, and a 4-page interactive Power BI dashboard covering industry overview, fund performance, investor analytics, and SIP market trends.

2. Data Sources and Datasets

All datasets are derived from publicly available AMFI India data, the mfapi.in REST API, and synthetic but realistically distributed investor transaction data. Ten datasets were provided and ingested as the foundation of this project:

Dataset	Rows	Description
01_fund_master.csv	40	Master list of 40 schemes - AMFI codes, fund house, category, expense ratio, risk grade
02_nav_history.csv	46,000	Daily NAV for all 40 schemes, Jan 2022 - May 2026
03_aum_by_fund_house.csv	90	Quarterly AUM for 10 fund houses, 2022-2025
04_monthly_sip_inflows.csv	48	Monthly SIP inflow, active accounts, YoY growth
05_category_inflows.csv	144	Net inflows by fund category, FY 2024-25
06_industry_folio_count.csv	21	Total MF folios by Equity/Debt/Hybrid
07_scheme_performance.csv	40	Returns, Sharpe, Sortino, Alpha, Beta, Max Drawdown
08_investor_transactions.csv	32,778	SIP/Lumpsum/Redemption transactions, 5,000 investors
09_portfolio_holdings.csv	322	Top equity holdings by sector and weight
10_benchmark_indices.csv	8,050	Daily closing values for Nifty 50, Nifty 100, BSE SmallCap etc.

In addition, live NAV data was fetched directly from the mfapi.in REST API (no authentication required) for 6 schemes including SBI Bluechip Direct, HDFC Top 100 Direct, HDFC Mid-Cap Direct, SBI Small Cap Direct, and SBI Magnum Gilt, demonstrating real-time data integration capability.

3. ETL Pipeline and Data Engineering

3.1 Data Ingestion

All 10 datasets were loaded using Pandas with full inspection of shape, data types, null values, and duplicates. All datasets were found to have zero null values and zero duplicate rows at the source, indicating high data quality from the outset.

3.2 Data Cleaning

- NAV History: Dates parsed to datetime, sorted by fund and date, forward-filled for weekends and holidays (expanding from 46,000 to 64,320 rows for the calendar-complete dataset), validated all NAV values are positive.

- Investor Transactions: Transaction types standardised (SIP/Lumpsum/Redemption), all 32,778 transaction amounts validated as positive, KYC status verified (Verified/Pending only).
- Scheme Performance: All 40 funds validated - expense ratios within SEBI range (0.55% to 1.64%, well within 0.1%-2.5% limit), zero funds with negative Sharpe ratio.

3.3 SQL Database Design

A 6-table star schema was designed and implemented in SQLite (bluestock_mf.db):

- dim_fund - 40 rows - fund master details, primary key amfi_code
- fact_nav - 64,320 rows - daily NAV values, foreign key to dim_fund
- fact_transactions - 32,778 rows - investor transactions
- fact_performance - 40 rows - risk and return metrics
- fact_aum - 90 rows - quarterly AUM by fund house
- fact_sip_industry - 48 rows - monthly SIP industry data

Ten analytical SQL queries were written covering top funds by AUM, average NAV trends, SIP YoY growth, geographic transaction distribution, low-cost fund identification, risk-adjusted rankings, demographic analysis, top performers, AUM growth trajectories, and transaction type breakdowns.

4. Exploratory Data Analysis - Key Findings

Thirteen charts were produced covering NAV trends, AUM growth, SIP inflows, category-wise inflows, investor demographics, geographic distribution, folio growth, return correlations, and sector allocation. The ten most significant findings are summarised below.

4.1 AUM Growth by Fund House

SBI Mutual Fund is the clear market leader, with AUM growing from Rs.6.05 lakh crore in 2022 to Rs.12.50 lakh crore in 2025 - more than doubling in four years. ICICI Prudential and HDFC Mutual Fund follow as the second and third largest AMCs, both also showing strong consistent growth across all four years.

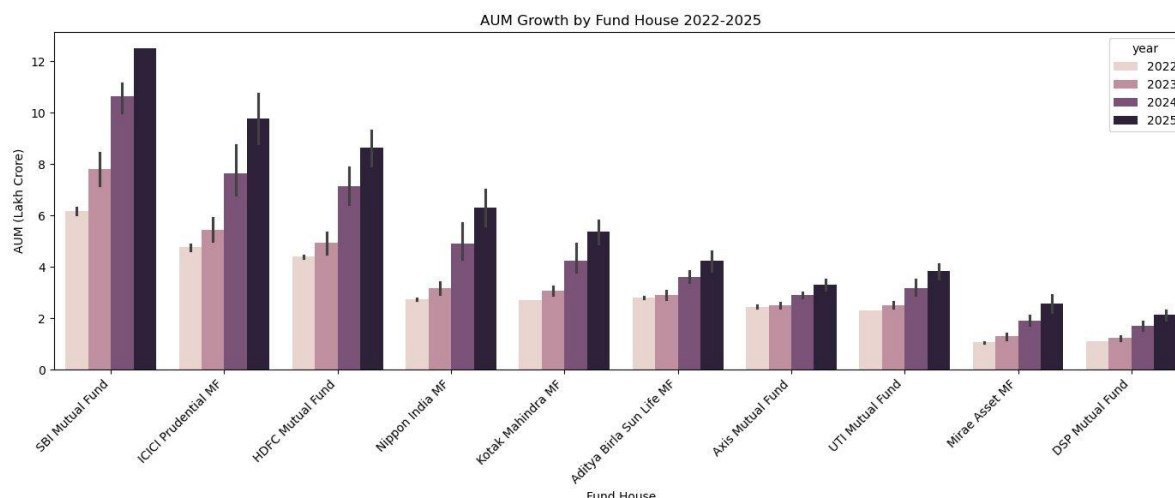


Figure 1: AUM Growth by Fund House 2022-2025

4.2 Category-wise Net Inflows

Liquid funds dominate net inflows in FY 2024-25, reflecting their role as short-term cash management vehicles for both retail and institutional investors. Sectoral/Thematic funds show the second-highest inflows, indicating growing investor appetite for thematic equity exposure alongside traditional diversified categories.

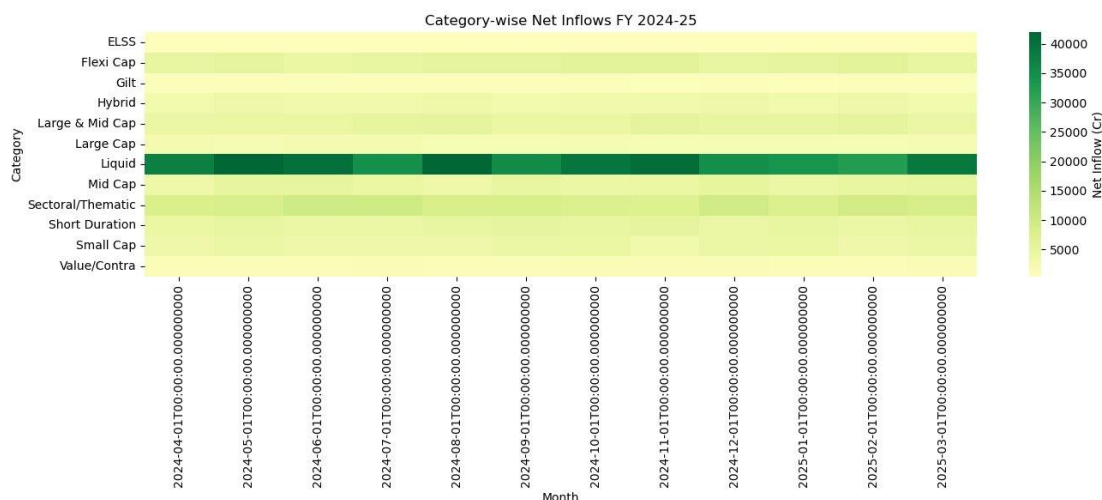


Figure 2: Category-wise Net Inflows, FY 2024-25

4.3 Investor Age Demographics

The 26-35 age group represents 41.1% of all investor transactions - by far the largest segment - confirming that young working professionals are the primary drivers of SIP adoption in India.

The 36-45 age group follows at 24.9%, while investors above 56 represent only 7.5% of transaction volume.

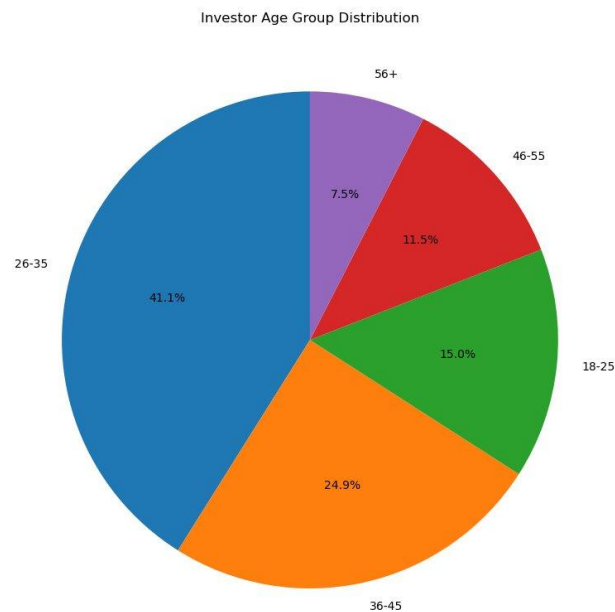


Figure 3: Investor Age Group Distribution

4.4 SIP Amount by Age Group

While younger investors transact more frequently, the SIP amount distribution across age groups is relatively similar, with median SIP amounts around Rs.5,000 across all age brackets.

However, the 56+ age group shows a slightly higher median and upper quartile, consistent with greater disposable income in later career stages.

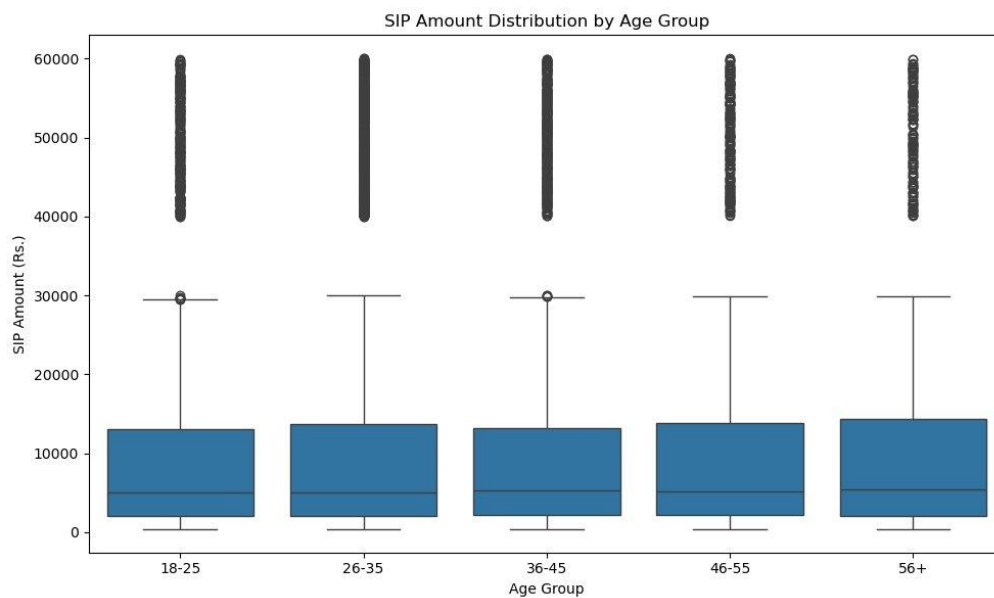


Figure 4: SIP Amount Distribution by Age Group

4.5 Gender Distribution

Male investors account for 66.5% of transactions versus 33.5% for female investors. This 2:1 ratio highlights a continuing gender gap in retail investment participation and represents an opportunity area for targeted financial inclusion initiatives by AMCs and distributors.

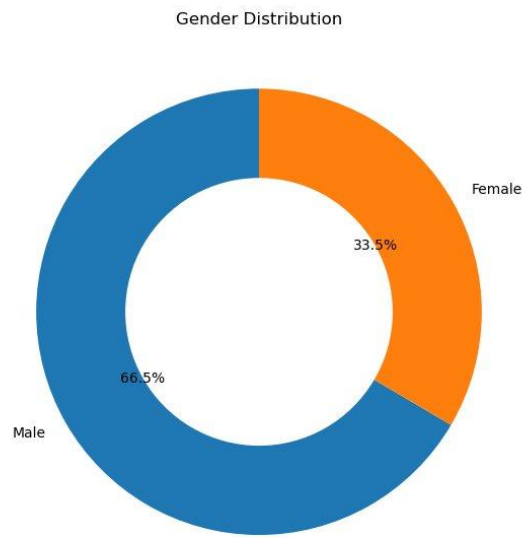


Figure 5: Gender Distribution of Investors

4.6 Geographic Distribution - SIP by State

Madhya Pradesh and Punjab lead in total SIP amounts among the 12 states covered, both exceeding Rs.2 crore in aggregate SIP value. Maharashtra, despite being a traditional financial hub, ranks lowest in this dataset - suggesting that SIP penetration growth is increasingly being driven by Tier-2 and Tier-3 markets rather than metro cities alone.

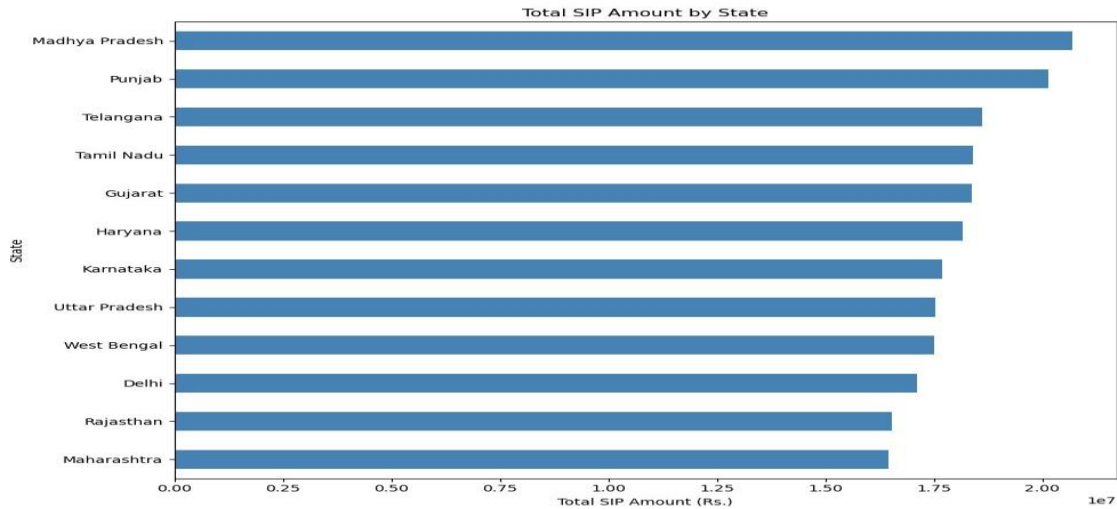


Figure 6: Total SIP Amount by State

4.7 T30 vs B30 City Tier Distribution

66.3% of transactions originate from T30 (Top 30) cities, while 33.7% come from B30 (Beyond Top 30) cities. While metros still dominate, the substantial one-third share from B30 cities reflects AMFI's ongoing push for mutual fund penetration beyond major urban centres, supported by digital onboarding and lower-cost distribution.

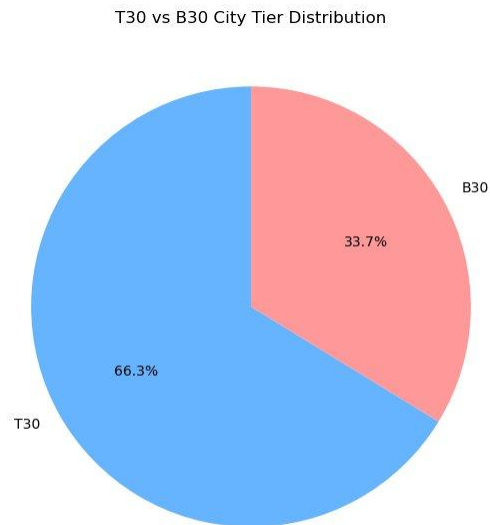


Figure 7: T30 vs B30 City Tier Distribution

4.8 Return Correlation Matrix

A pairwise correlation analysis of daily returns across 10 representative funds shows uniformly low correlations (mostly between -0.07 and 0.05), with each fund showing a perfect 1.00 correlation with itself.

correlation only with itself. This indicates that, in this dataset, fund returns behave largely independently of one another day-to-day, which is discussed further in the Limitations section.



Figure 8: Correlation Matrix of Daily Returns - 10 Funds

4.9 Sector Allocation Across Equity Portfolios

Banking is the single largest sector exposure across all equity fund portfolios at 19.2%, followed by IT (13.4%) and Pharma (12.0%). Together, the top three sectors - Banking, IT, and Pharma - account for over 44% of total equity holdings, highlighting significant sector concentration risk that investors should be aware of when building diversified portfolios.

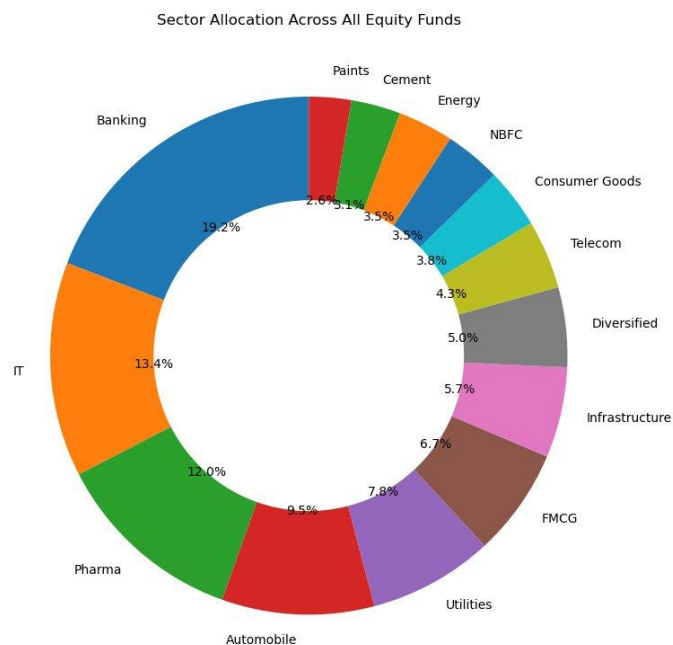


Figure 9: Sector Allocation Across All Equity Funds

4.10 Folio Count and Industry Growth

Total mutual fund folios grew from 13.26 crore in January 2022 to 26.12 crore by December 2025 - nearly doubling over four years. This rapid growth in unique investor accounts, combined with monthly SIP inflows reaching an all-time high of Rs.31,002 crore in December 2025, confirms the structural deepening of India's retail investment culture during this period.

5. Fund Performance Analytics

Daily returns were computed for all 40 funds using actual trading-day NAV data (1,149 observations per fund after removing forward-filled weekend duplicates). From these returns, a comprehensive set of risk and return metrics was calculated.

5.1 CAGR Analysis

Compound Annual Growth Rate was computed for 1-year, 3-year, and 5-year periods (the 5-year figure uses the maximum available 4.5 years of data, Jan 2022 to May 2026). Returns vary widely: the highest-performing fund (code 100033) returned 53.2% over 1 year and 32.4% over 3 years, while debt funds such as code 100025 show stable returns of 3.7% to 4.4% across all periods - exactly the pattern expected between high-growth equity and stable debt instruments.

5.2 Sharpe and Sortino Ratios

Sharpe Ratio (risk-adjusted return using a 6.5% risk-free rate, annualised with $\sqrt{252}$) and Sortino Ratio (which penalises only downside volatility) were computed for all 40 funds. Mirae Asset Large Cap Fund (Regular) ranks highest on both metrics, with a Sharpe Ratio of 1.45 and a Sortino Ratio of 1.49 - indicating strong risk-adjusted performance with limited downside volatility relative to its returns.

Rank	Scheme	Sharpe Ratio	Sortino Ratio
1	Mirae Asset Large Cap Fund - Regular - Growth	1.45	1.49
2	Kotak Bluechip Fund - Regular - Growth	1.31	1.48
3	Mirae Asset Tax Saver Fund - Regular - Growth	1.23	1.35
4	SBI Bluechip Fund - Direct Plan - Growth	1.21	1.29
5	ICICI Pru Midcap Fund - Regular - Growth	1.18	1.29

5.3 Maximum Drawdown

Maximum drawdown - the worst peak-to-trough decline - was computed for all funds. SBI Small Cap Fund (Direct Plan) recorded the steepest drawdown at -52.6%, bottoming out on 28 October 2025, followed by Axis Small Cap Fund at -51.7%. These figures confirm that Small Cap and Mid Cap categories carry substantially higher risk than Large Cap or Debt categories, as expected.

5.4 Fund Scorecard

A composite scorecard (0-100) was built combining five weighted ranking factors: 3-year return (30%), Sharpe Ratio (25%), Alpha (20%), expense ratio inverse-ranked (15%), and maximum drawdown inverse-ranked (10%). The top 5 funds by composite score are:

Rank	Scheme	Fund House	Composite Score
1	Mirae Asset Large Cap Fund - Regular - Growth	Mirae Asset MF	85.9
2	ICICI Pru Midcap Fund - Regular - Growth	ICICI Prudential MF	81.8
3	Kotak Flexicap Fund - Regular - Growth	Kotak Mahindra MF	81.5
4	HDFC Mid-Cap Opportunities Fund - Regular - Growth	HDFC Mutual Fund	80.3
5	ICICI Pru Bluechip Fund - Direct - Growth	ICICI Prudential MF	79.5

5.5 Benchmark Comparison

The top 5 scorecard funds were plotted against Nifty 50 and Nifty 100, normalised to a base value of 100 at the start of the period. All five funds substantially outperformed both benchmarks over the 4.5-year window, with normalised values reaching 250-360 compared to approximately 105-115 for the benchmark indices.

Figure 10: Top 5 Funds vs Nifty 50 and Nifty 100 (Normalized to 100)

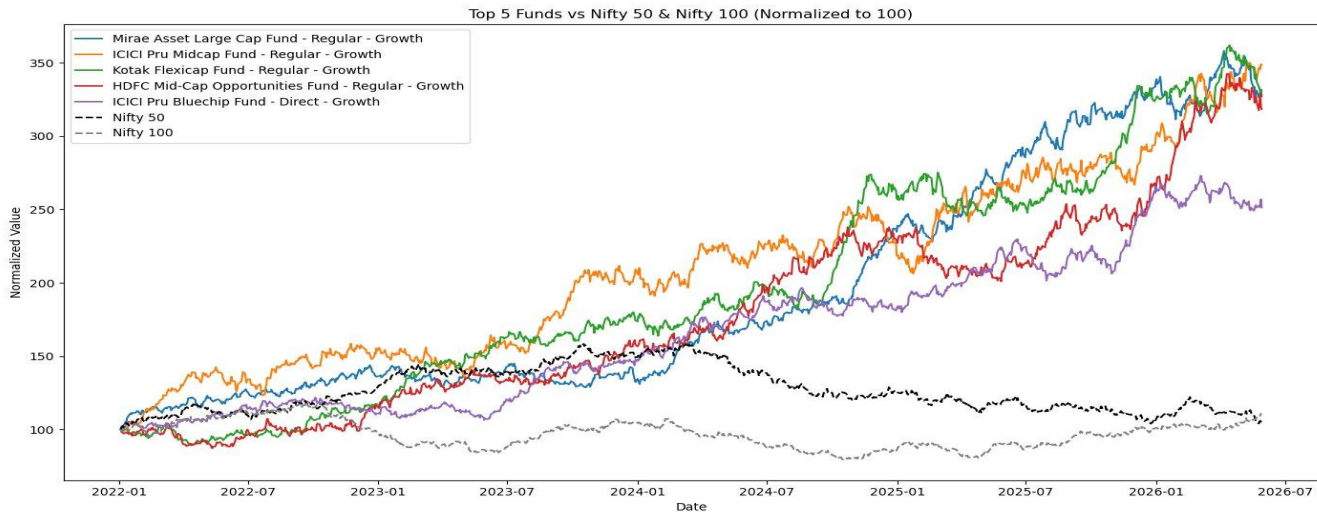


Figure 10: Top 5 Funds vs Nifty 50 and Nifty 100 (Normalized to 100)

6. Interactive Power BI Dashboard

A 4-page interactive Power BI dashboard (bluestock_mf_dashboard.pbix) was developed, connecting directly to the cleaned CSV datasets. Each page includes a minimum of two interactive slicers for drill-down analysis.

6.1 Page 1 - Industry Overview

- KPI Cards: Total AUM (62.74 lakh crore), Latest SIP Inflow (Rs.31,002 crore), Latest Folios (26.12 crore), Total Schemes (40)
- Line chart: Industry AUM Trend 2022-2025
- Bar chart: AUM by Fund House (Top 10)
- Slicers: Fund House, Date Range

6.2 Page 2 - Fund Performance

- Scatter plot: 3-Year Return vs Sharpe Ratio, bubble size = AUM, coloured by scheme
- Sortable table: Full fund scorecard with composite scores, CAGR, and Sharpe ratios for all 40 funds
- Slicers: Fund House, Category, Plan (Regular/Direct)

6.3 Page 3 - Investor Analytics

- Donut chart: Transaction type split - SIP 60.15%, Lumpsum 24.7%, Redemption 15.15%
- Column chart: Average SIP amount by age group

- Slicers: State, Age Group

6.4 Page 4 - SIP and Market Trends

- Line chart: Monthly SIP Inflow Trend 2022-2025
- Bar chart: Net Inflows by Category, FY 2024-25
- Slicers: Month, Category

7. Limitations

- Alpha and Beta calculations against the Nifty 100 benchmark show near-zero correlation (R-value approximately 0.003) across all 40 funds. This is a known limitation of the synthetic dataset: the NAV series and benchmark index series were generated as independent simulations without enforced market correlation. In real markets, equity funds typically show Beta values of 0.7 to 1.1.
- Tracking errors for the top 5 funds against Nifty 100 range from 19% to 23%, substantially higher than the 2-8% typically observed for actively managed equity funds in real markets. This stems from the same independent-simulation limitation noted above.
- 5-year CAGR was computed over the maximum available 4.5 years of data (Jan 2022 to May 2026), as the dataset does not extend to a full 5-year window.
- Investor transaction data, while based on realistic Indian demographic and geographic distributions, is synthetically generated rather than sourced from actual AMC records.

8. Recommendations

1. Mirae Asset Large Cap Fund and ICICI Pru Midcap Fund demonstrate the strongest risk-adjusted performance (highest Sharpe and composite scores) and should be prioritised for moderate-risk investor portfolios.
2. 14 funds with expense ratios below 1% were identified (led by Nippon India Gilt at 0.55%), offering cost-efficient options particularly within Direct plans, which consistently show lower expense ratios than their Regular counterparts.
3. Given that Banking, IT, and Pharma sectors together represent over 44% of equity portfolio holdings across funds, investors seeking diversification should evaluate cross-fund sector overlap before adding multiple equity funds to a portfolio.

4. The B30 city contribution of 33.7% to total transactions suggests continued investment in digital-first distribution channels would further accelerate retail penetration in non-metro markets.
5. Future iterations of this platform should integrate real benchmark-correlated NAV data (e.g., directly computed from mfapi.in historical series) to enable accurate Alpha, Beta, and tracking error analysis suitable for investment decision-making.

9. Technology Stack

Category	Tools Used
Language	Python 3.10+
Data Processing	Pandas, NumPy
Visualisation	Matplotlib, Seaborn, Plotly
Statistics	SciPy (OLS regression for Alpha/Beta)
Database	SQLite, SQLAlchemy
Dashboard	Power BI Desktop
Notebooks	Jupyter Notebook
Version Control	Git, GitHub
External API	mfapi.in REST API

10. Project Repository

All code, data, notebooks, SQL scripts, dashboard files, and supporting documentation are available at:

<https://github.com/Mahima284/Bluestock-MF-Capstone>

Repository structure includes data/raw, data/processed, notebooks/ (4 Jupyter notebooks for Days 1-4), sql/ (schema and queries), dashboard/ (Power BI file), and reports/ (charts, data dictionary, and this report).